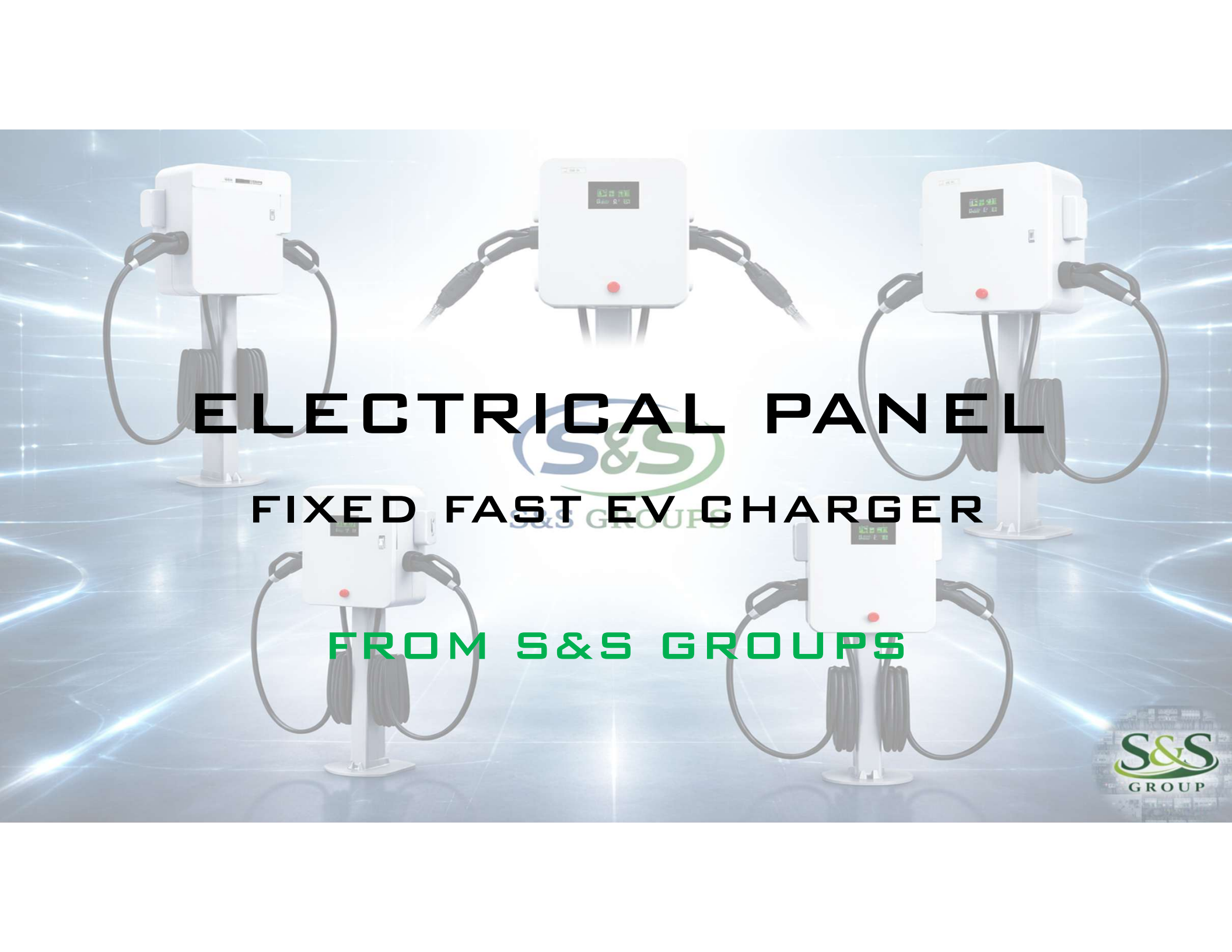




ELECTRICAL PANEL

FIXED FAST EV CHARGER

FROM S&S GROUPS

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Fixed Fast EV Charger

Description About Fixed Fast EV Charger

➤ The **Fixed Fast EV Charger** is a high-efficiency electric vehicle charging system designed for fast and reliable charging of electric vehicles equipped with lithium-ion battery packs. The system is engineered for safe operation, high performance, and long service life, making it suitable for commercial charging stations, parking facilities, and residential charging applications.

- Panel Construction
- Busbar System (Aluminium Type)
- Incoming Supply
- Outgoing Feeders
- Protection System
- Metering & Indication
- Internal Wiring
- Cable Entry
- Earthing
- Testing & Inspection
- Documentation
- Scope of Supply & Warranty

Robust IP55-Rated Casing

Robust IP55-Rated Casing
Robust Heavy-Duty Construction

DC Output Contactor + MCB

Smart EV Controller
DC 50 Amp

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Fixed Fast EV Charger

1. PANEL CONSTRUCTION

- ❑ The EV Charging Panel shall be fabricated from **1.6 mm thick CRCA sheet steel** for structure, properly reinforced with stiffeners.
- ❑ The panel shall be **floor mounted, front operated, cubicle type**, dust and vermin proof construction, finished with **7 tank anti-corrosive treatment and powder coated with epoxy polyester paint (minimum 70–80-micron thickness)**.

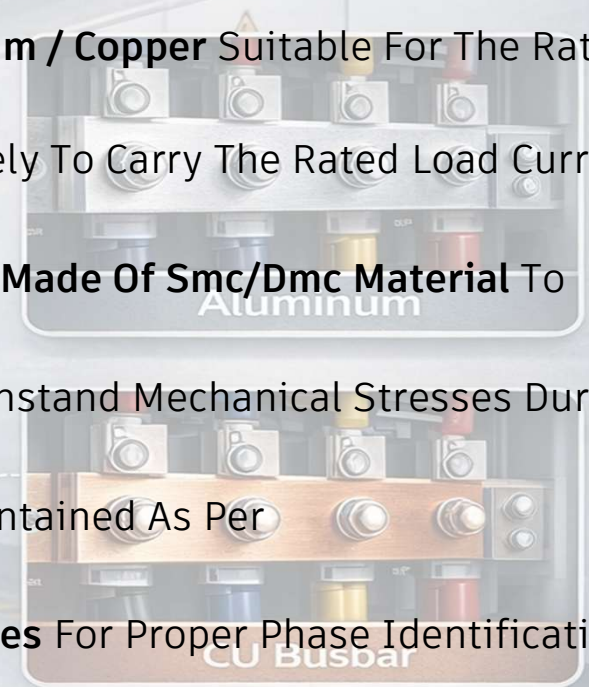
➤ Body Construction :-

- The enclosure of the Fixed Fast EV Charger shall be fabricated from high-quality **CRCA sheet steel of minimum 1.6 mm thickness**, precision sheet metal processed, folded and welded to form a **robust, dust and weather-resistant structure** suitable for outdoor installation. The panel shall be **floor-mounted / pedestal type construction** with proper reinforcement for mechanical strength and stability. The enclosure shall be provided with **hinged lockable front doors with neoprene rubber gaskets** to ensure protection against dust and moisture, maintaining minimum **IP54 protection level**. The surface of the panel shall undergo **seven-tank pretreatment process followed by electrostatic powder coating (minimum 70–80 microns thickness)** in standard shade for corrosion resistance and long service life. Adequate **ventilation louvers with wire mesh**, cable entry provisions from bottom/top with gland plates, and proper earthing studs shall be provided for safe and reliable operation.

Fixed Fast EV Charger

2. BUSBAR (ALUMINUM/COPPER TYPE)

- The Main Busbars Shall Be Made Of **High Conductivity Grade Aluminum / Copper** Suitable For The Rated Current Of The EV Charging Panel.
- Busbars Shall Be **Rectangular Cross-section Type** And Sized Adequately To Carry The Rated Load Current Without Excessive Temperature Rise.
- The Busbars Shall Be **Mounted On High Strength Insulated Supports Made Of Smc/Dmc Material** To Provide Mechanical Strength And Electrical Insulation.
- Busbar Supports Shall Be **Fixed Firmly On The Panel Structure** To Withstand Mechanical Stresses During Operation And Short Circuit Conditions.
- Adequate **Phase To Phase And Phase To Earth Clearance** Shall Be Maintained As Per **Relevant Is / Iec Standards**.
- Busbars Shall Be Provided With **Heat Shrinkable Pvc Insulation Sleeves** For Proper Phase Identification.
- **Colour Coding Of Busbars Shall Be Provided As Per Standards:**
 - ✓ R Phase – Red
 - ✓ Y Phase – Yellow
 - ✓ B Phase – Blue
 - ✓ Neutral – Black
- Busbars shall be arranged in a **compact and accessible manner** for easy maintenance and inspection.



ALM BUSBAR



CU BUSBAR

3. INPUT POWER SUPPLY

- ❑ The charger operates on a wide input voltage range to ensure stable performance even During voltage fluctuations.

- **Input Voltage Range:** 160 VAC – 270 VAC

- **Operating Frequency:** 50 Hz

- **Rated Input Voltage:** 240 VAC

- ❑ This wide input range ensures reliable operation in varying power supply conditions.

4. BATTERY COMPATIBILITY

- ❑ The charger is designed to support multiple lithium-ion battery configurations commonly used in electric vehicles :-

- **Battery Type:** Lithium-Ion

- **Supported Battery Voltages:**

- ✓ 48 V
- ✓ 60 V
- ✓ 72 V

- **Battery Capacity Range:** 27 Ah – 100 Ah

- ❑ This flexibility allows compatibility with a wide range of electric two-wheelers and light EVs.

2 Battery Compatibility

⚡ Battery Type

• Lithium-Ion

⚡ Supported Battery Voltages

• 48 V

• 60 V

• 72 V

⚡ Battery Capacity Range

27 Ah – 100 Ah



Battery Capacity Range

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5. CHARGING VOLTAGE

Charging Voltage | Output Current | Charging Efficiency

❑ The charger provides a regulated DC output suitable for efficient battery charging :-

➤ **Charging Voltage Range:** 36 V – 84 V DC

❑ The system automatically adjusts voltage according to the connected battery pack.

6. OUTPUT CURRENT

❑ The charger maintains precise output current control to ensure safe and efficient battery charging :-

➤ **Output Current Accuracy:** $\pm 3\%$

❑ This ensures battery protection and consistent charging performance.

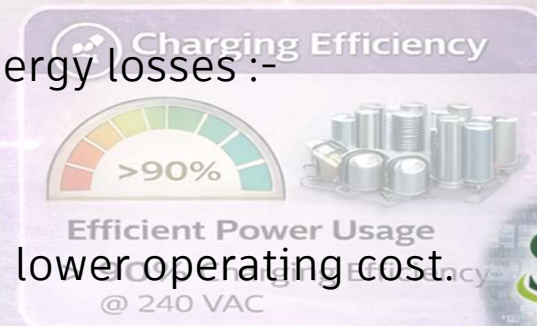
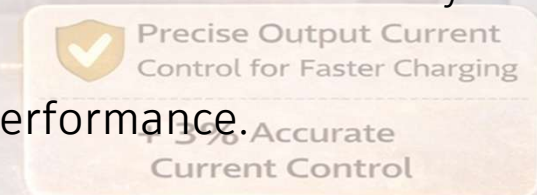
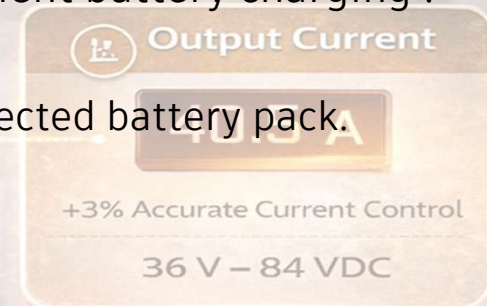
7. CHARGING EFFICIENCY

❑ The charger is designed for high efficiency to minimize energy losses :-

➤ **Charging Efficiency:** Greater than 90 %

➤ **Reference Input Voltage:** 240 VAC

❑ High efficiency results in reduced power consumption and lower operating cost.



8. POWER PLUG TYPE

Power Plug Type, Charging Cable & Application View

❑ The charging unit uses a high-capacity DC output connection suitable for fast charging :-

- **Power Rating:** DC 50 Amp

9. CHARGING CABLE

❑ The charger is equipped with a durable charging cable suitable for outdoor charging use :-

- **CONNECTOR TYPE:** TYPE-6 CONNECTOR
- **CABLE LENGTH:** 2.5 METER

10. APPLICATION

❑ This fast charger is suitable for :-

- EV charging stations
- Parking areas
- Residential charging points
- Commercial EV infrastructure

Application



EV Charging Stations



Parking Areas



Residential Charging Points



EV Charging Stations



Parking Areas



Residential Charging Points



Commercial EV Infrastructure

Power Plug Type



DC 50 Amp

Robust DC power plug capable of delivering up to 50 Amps for fast



11. DISPLAY & MONITORING SYSTEM

Display & Monitoring System & Key Features

- ❑ The charger includes an integrated digital display for monitoring charging parameters and operational status :-

➤ THE DISPLAY SHOWS:

- ✓ Charging parameters
- ✓ Real-time SOC (State of Charge)
- ✓ Discharge count
- ✓ Battery temperature
- ✓ Gun temperature
- ✓ Charger status
- ✓ Payment balance deduction

12. KEY FEATURES

- Fast charging capability
- Wide input voltage range
- Lithium-ion battery compatibility
- High charging efficiency (>90%)
- Digital monitoring display
- Safe and reliable operation
- Compact and robust design

Key Features

-  Fast Charging Capability
-  High Charging Efficiency
-  Wide Input Voltage Range
-  Compatibility with Li-Ion Battery Packs
-  Dual Type-6 Connectors
-  Digital Monitoring Display
-  IP55 Rated Casing
-  Compact & Robust Design
-  Safe & Reliable Operation

Fixed Fast EV Charger – Internal Wiring & Cable Entry View

13. INTERNAL WIRING

- INTERNAL WIRING SHALL BE DONE USING **FRLS COPPER CONTROL CABLES** OF **ADEQUATE RATING**.
- ALL WIRES SHALL BE **PROPERLY FERRULED** AT BOTH ENDS FOR IDENTIFICATION.
- WIRING SHALL BE **NEATLY ROUTED** THROUGH **PVC CHANNELS / CABLE DUCTS**.
- PROPER **TERMINATION** USING **CRIMPING LUGS** AND **TERMINAL BLOCKS** SHALL BE PROVIDED.

Wiring Color Code

—	L – Brown
—	N – Blue
—	E – Green/Yellow
—	DC + – Red
—	DC – – Black
—	CAN-H – Green
—	CAN-L – White

Door Earth Connection

14. CABLE ENTRY

- CABLE ENTRY SHALL BE PROVIDED FROM **BOTTOM/TOP** OF THE PANEL THROUGH **REMOVABLE GLAND PLATES**.
- **SUITABLE DOUBLE COMPRESSION CABLE GLANDS** SHALL BE USED FOR CABLE TERMINATION.
- **GLAND PLATE** SHALL BE **PROPERLY EARTHED** TO THE PANEL BODY.
- **ADEQUATE SPACE** SHALL BE PROVIDED FOR **SAFE CABLE BENDING** AND **TERMINATION**.

Internal Layout

- Neat Wiring & Labeling
- DIN Rail Mounting
- Cable Glands
- Earth Bonding

Fixed Fast EV Charger – Earthing & Testing & Inspection View

15. EARTHING

- **PANEL BODY** – PROPERLY CONNECTED TO EARTH USING GI/COPPER EARTH WIRE
- **DOOR EARTHING** – FLEXIBLE COPPER STRIP PROVIDED BETWEEN DOOR AND PANEL BODY
- **GLAND PLATE** – PROPERLY EARTHED TO THE MAIN PANEL EARTH POINT
- **EARTH STUD** – DEDICATED EARTH STUD PROVIDED INSIDE THE PANEL FOR SAFE EARTHING CONNECTION

16. TESTING & INSPECTION

- **Visual Inspection** – Checking of panel structure, wiring and component mounting
- **Insulation Resistance Test** – IR test performed using 500V/1000V megger
- **Continuity Test** – Verification of wiring and earthing continuity
- **Functional Test** – Complete operation of panel checked before dispatch

- #### Testing & Inspection
- ✓ Earth Continuity Test
 - ✓ Input Voltage Measurement
 - ✓ Output Voltage & Current Test
 - ✓ Insulation Resistance Test
 - ✓ Ground Fault Protection Test
 - ✓ Functional Test of Components
 - ✓ Display Verification

- #### Earthing Details
- ✓ Earth Wiring Continuity
 - ✓ Labels & Bonding

- ✓ Earth Rod Resistance < 5 Ω
- ✓ Earth Ronkaging

Fixed Fast EV Charger Documentation & Scope Includes & Warranty

➤ The supplier shall provide :-

- ✓ User Manual
- ✓ GA drawing
- ✓ Data sheet & Brochure
- ✓ SLD diagram (Single Line Drawing)
- ✓ Test Reports
- ✓ Wiring Diagram
- ✓ Test certificates
- ✓ Operation & maintenance manual
- ✓ Warranty

Scope Includes

- ✓ Fixed Fast EV Charger
- ✓ Mounting Stand
- ✓ Charging Cable 2.5 Meter (Type 6 Connector)
- ✓ User Manual
- ✓ Documentation
- ✓ Test Reports & Certificates
- ✓ AC Input Cable (Without Plug)

Warranty

- ✓ 1 Year Standard Warranty
- ✓ Service Support Included
- ✓ Extended Warranty Options Available

18. SCOPE INCLUDES & WARRANTY

➤ THE SCOPE SHALL INCLUDE :-

- ✓ Design & engineering
- ✓ Fabrication & Assembly
- ✓ Supply of all Components
- ✓ Internal wiring and Bus bar
- ✓ Testing & quality inspection
- ✓ Packing and transportation

- ✓ Covers all manufacturing

WARRANTY :-

Minimum **12 months warranty** from date of commissioning.

Support

- Includes technical support & service support

Options

- ✓ Extended Warranty of 1 or 2 Years
- ✓ Covers within 3-4 weeks